

MALACCA, Malaysia

WMO: 486650

Lat: 2.2551N

Lon: 102.2440E

Elev: 11

StdP: 101.20

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	22.8	23.1	19.8	14.5	28.4	20.4	15.1	27.8	6.8	29.4	6.2	29.5	1.6	40	0.330

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	7.2	34.0	26.1	33.2	26.2	32.9	26.2	27.7	31.7	27.5	31.4	27.2	31.1	3.3	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.7	22.3	30.3	26.2	21.7	29.9	26.1	21.5	29.8	88.4	32.1	87.2	31.9	86.1	31.3	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.0	5.2	4.5	DB	21.5	35.3	0.9	1.2	20.9	36.2	20.3	36.8	19.8	37.5	19.2	38.3	(4)
(5)			WB	20.3	28.6	1.1	0.8	19.5	29.2	18.9	29.7	18.3	30.2	17.5	30.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.0	27.5	28.2	28.6	28.6	28.5	28.2	27.7	27.7	27.8	28.0	27.7	27.4
	DBStd	1.00	1.07	1.05	1.05	0.95	0.86	0.87	0.88	0.85	0.80	0.83	0.77	0.98
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6568	543	510	576	558	574	546	550	548	533	557	531	541
	CDD18.3	3526	285	277	317	308	316	296	292	290	283	299	281	282
	CDH23.3	36664	2651	2790	3356	3370	3644	3347	3162	3083	2976	3085	2646	2555
	CDH26.7	13833	921	1108	1373	1366	1491	1322	1177	1121	1092	1151	891	819
Wind	WSAvg	1.8	2.7	2.8	2.1	1.6	1.5	1.5	1.5	1.6	1.5	1.6	1.6	2.1
Precipitation	PrecAvg	1945	89	86	138	181	169	157	184	195	173	201	231	158
	PrecMax	2619	271	279	361	391	427	466	472	491	367	347	424	492
	PrecMin	1169	4	0	7	13	25	0	50	64	20	58	99	16
	PrecStd	268	65	69	73	80	78	94	83	91	71	78	77	89
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.2	34.7	35.0	34.8	34.1	33.2	33.0	32.9	33.1	33.2	33.0	33.0
		MCWB	25.5	25.1	25.4	26.9	27.0	26.5	26.3	26.4	26.1	26.0	26.3	26.1
	2%	DB	32.5	33.7	33.9	33.4	33.2	32.8	32.2	32.1	32.2	32.7	32.2	32.1
		MCWB	25.4	25.3	25.8	26.7	26.8	26.5	26.2	26.2	26.1	26.1	26.2	25.9
	5%	DB	31.8	32.9	33.0	32.8	32.7	32.1	31.8	31.7	31.9	32.0	31.6	31.3
		MCWB	25.3	25.4	26.1	26.7	26.7	26.4	26.1	26.1	26.1	26.1	26.2	25.8
	10%	DB	31.0	32.0	32.1	32.0	32.0	31.7	31.1	31.0	31.1	31.2	30.9	30.7
		MCWB	25.2	25.3	26.0	26.6	26.6	26.4	26.1	26.1	26.1	26.0	26.2	25.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.5	27.4	27.7	28.1	28.1	27.8	27.6	27.8	27.5	27.5	27.6	27.4
		MCDB	31.4	32.0	32.2	32.6	32.3	31.4	31.1	31.1	31.3	31.7	31.5	31.5
	2%	WB	26.6	26.7	27.1	27.6	27.6	27.3	27.1	27.0	27.0	26.9	27.0	26.7
		MCDB	30.6	31.1	31.6	31.9	31.6	31.0	30.6	30.7	30.8	30.9	30.9	30.5
	5%	WB	26.1	26.2	26.7	27.2	27.2	27.0	26.7	26.6	26.6	26.6	26.6	26.3
		MCDB	30.0	30.7	31.0	31.3	31.1	30.7	30.2	30.2	30.3	30.5	30.3	29.9
	10%	WB	25.6	25.9	26.4	26.7	26.9	26.7	26.4	26.3	26.3	26.2	26.2	25.9
		MCDB	29.4	30.3	30.6	30.7	30.7	30.2	29.8	29.7	29.8	30.0	29.8	29.4
Mean Daily Temperature Range	5% DB	MDBR	7.2	8.1	8.1	7.6	7.2	7.0	6.9	6.9	7.1	7.3	7.1	6.7
		MCDBR	8.3	9.2	8.9	8.4	8.0	7.6	7.6	7.5	7.8	8.0	7.9	7.7
		MCWBR	3.0	3.0	3.1	3.2	3.1	2.9	2.9	3.1	3.1	3.0	3.1	3.0
	5% WB	MCDBR	7.5	8.2	8.1	7.8	7.4	7.1	7.0	7.0	7.3	7.5	7.4	7.2
MCWBR		3.2	3.3	3.3	3.3	3.2	3.1	3.1	3.2	3.2	3.2	3.3	3.1	
Clear-Sky Solar Irradiance	taub	0.470	0.511	0.528	0.505	0.488	0.522	0.503	0.541	0.562	0.587	0.479	0.455	
	taud	2.277	2.134	2.099	2.174	2.224	2.122	2.183	2.063	2.004	1.924	2.254	2.332	
	Ebn at Noon	859	830	812	814	805	765	786	774	774	758	844	867	
	Edh at Noon	139	163	169	153	141	153	145	169	183	199	141	130	
All-Sky Solar Radiation	RadAvg	4.61	5.27	5.28	5.17	4.89	4.70	4.65	4.74	4.69	4.70	4.50	4.26	
	RadStd	0.40	0.41	0.37	0.22	0.20	0.22	0.23	0.20	0.33	0.35	0.25	0.28	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.34	N/A	N/A	N/A	+0.30	+0.30	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	+0.40	+0.18	N/A	+0.44	+0.51	+0.44	N/A	N/A	+153	+153

Nomenclature: See separate page